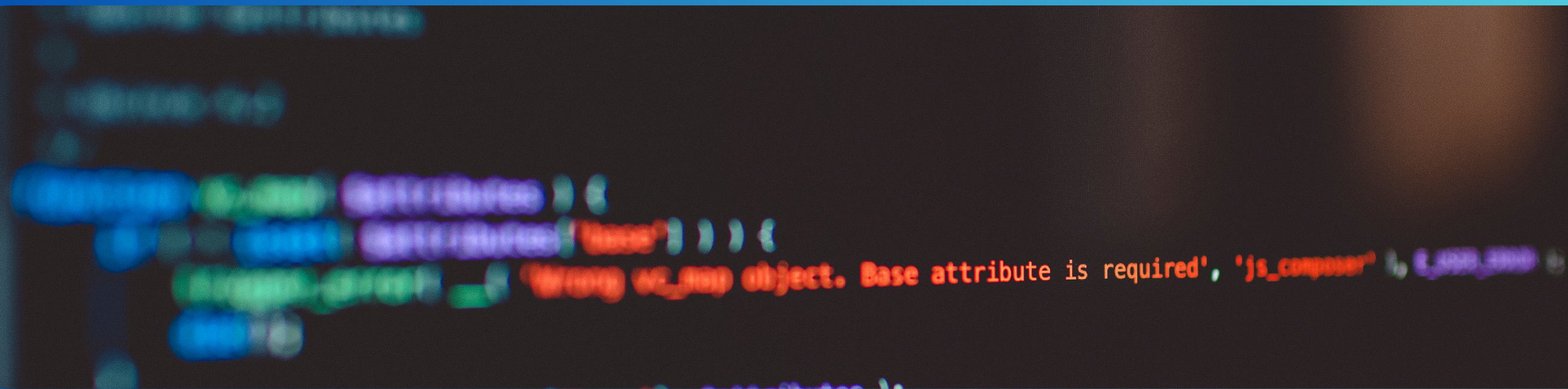


EternalBlue

Network Security

Attack Objectives

- Gain unauthorized access
- Data theft or manipulation
- Elevate privileges



The image shows a blurred screenshot of a code editor. The visible text includes a JavaScript error message: "Wrong vm_map object. Base attribute is required", "js_composer". The code is written in a dark theme with syntax highlighting.

Network Environment

- Platform: TryHackMe VMs
- Network state: Isolated lab, no public internet egress (TryHackMe private network)
- Victim: Unpatched Windows 7 VM SMBv1 enabled, SMB reachable on TCP/445
- Attacker: TryHackMe AttackBox (Ubuntu-based) used Metasploit to run the EternalBlue exploit



What is EternalBlue

EternalBlue is a Windows exploit developed by the NSA and leaked by the Shadow Brokers in 2017. It targets a flaw in Microsoft's Server Message Block (SMB) protocol, which is used for sharing files, printers, and data between computers on a network, allowing attackers to send malicious data and gain control of vulnerable .systems



Attack severity

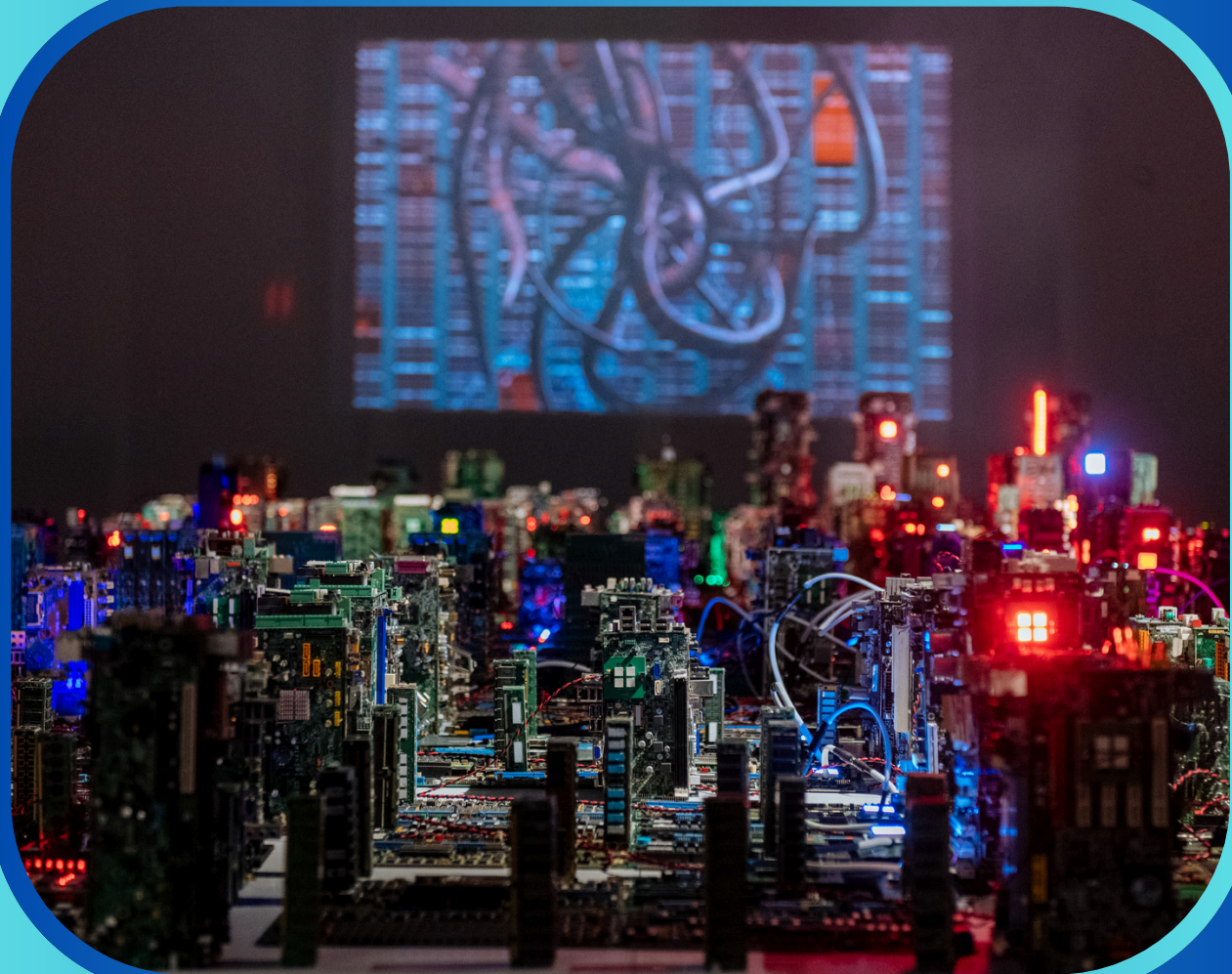
Remote Code Execution (RCE)

High / Critical severity (requires immediate patching)

Lead to full system compromise

Cause data theft, file destruction, or service disruption

Eternalblue target



Old/unpatched Windows

unpatched server with SMBv1 enabled
(SMB reachable on TCP port 445)

Attack prevention



Apply MS17-010 / Windows security updates immediately

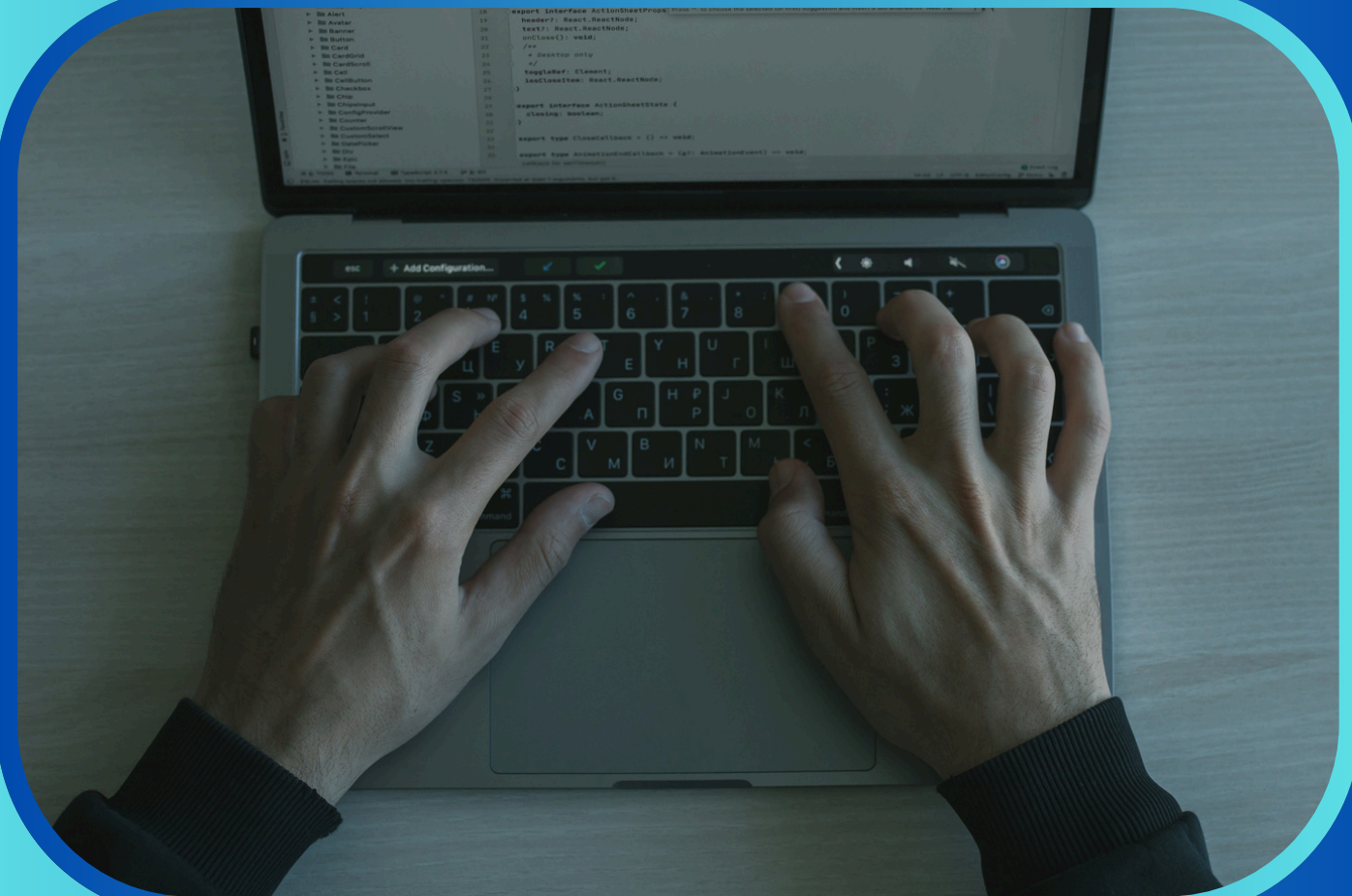
Disable SMBv1

Block/limit inbound SMB (port 445) at perimeter

Network segmentation + IDS monitoring

Regular backups and user awareness training

EternalBlue tool

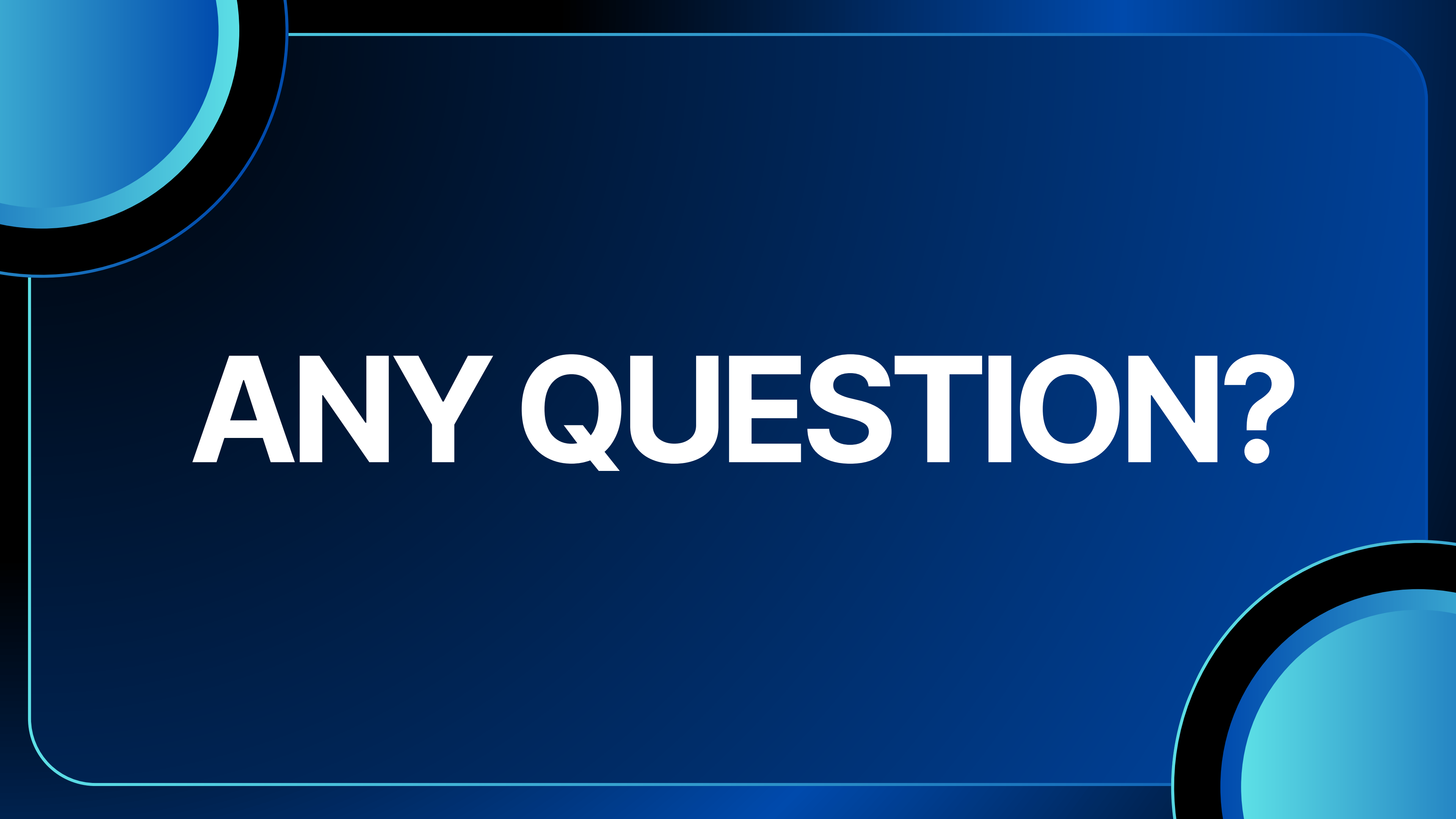


TryHackMe — Blue room

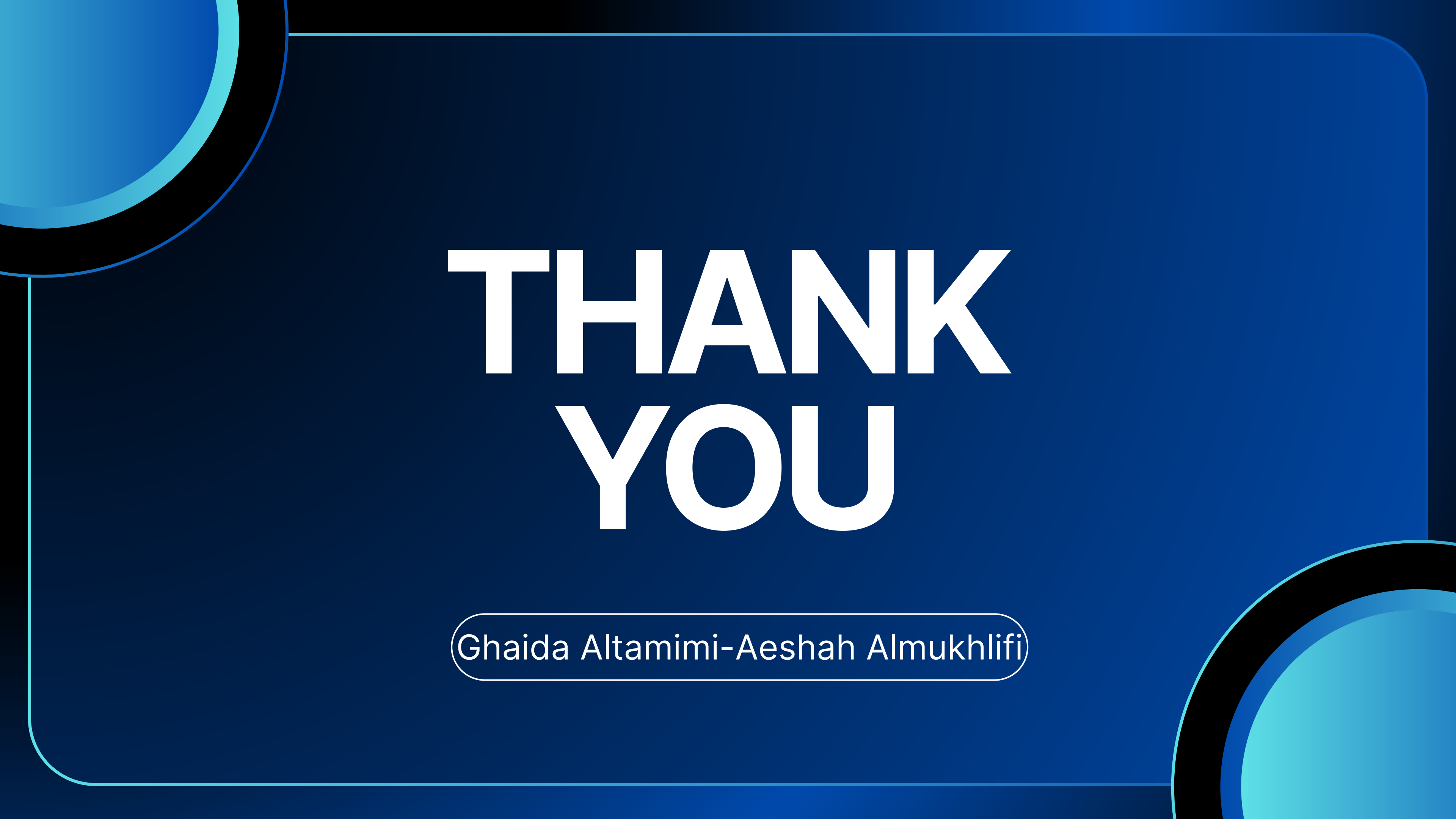
All interaction was carried out inside the AttackBox



DEMO



ANY QUESTION?



THANK YOU

Ghaida Altamimi-Aeshah Almukhlifi